

The two-row $d = 4$ fiber law for $b \equiv 2, 3 \pmod{4}$:
a central-binomial closed form, the Gegenbauer engine,
and why the obstruction is arithmetic

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Setup. For $\lambda = (2m - b, b)$, $s = 1 + i$, $A = 1 + su + u^2$, write $g_b(m) = [u^b]A^m$, $G_b(m) = [u^b]((1 - u)A^m) = g_b(m) - g_{b-1}(m)$, and $I_b(m) = \text{Im } G_b(m) \in \mathbb{Z}[m]$. After dividing out the forced roots $m = 0, \dots, R$ ($R = \lfloor (b-1)/2 \rfloor$), the two-row $d = 4$ law $G_{(2m-b,b)}(i) = 0 \iff (2, 2)$ is the statement

$$(\star) \quad I_b(m) \neq 0 \text{ for every integer } m \geq b, \quad b \equiv 2, 3 \pmod{4}.$$

The classes $b \equiv 0, 1$ are proved. For $b \equiv 2, 3$, $I_b(m) \equiv m \binom{m-1}{R} \pmod{2}$: the bare factor m rules out every single-prime certificate.

1 The engine

Proposition 1. For all $b \geq 2$, $b g_b(m) = s(m - b + 1)g_{b-1}(m) + (2m - b + 2)g_{b-2}(m)$, with $g_0 = 1$, $g_1 = sm$.

Proof. From $A(A^m)' = m(s + 2u)A^m$ read off $[u^{b-1}]$. □

Proposition 2. $g_b(m) = \binom{m}{b} s^b {}_2F_1\left(-\frac{b}{2}, -\frac{b-1}{2}; m - b + 1; -2i\right) = C_b^{(-m)}\left(-\frac{s}{2}\right)$, the Gegenbauer polynomial of degree b in the parameter $-m$ at $x = -s/2$.

Because the recurrence coefficient $(2m - b + 2)/b$ depends on m , the $g_b(m)$ are *not* orthogonal polynomials in m ; the classical zero-distribution theory does not apply directly.

2 The half-integer value

Theorem 1. For all $b \geq 1$,

$$G_b\left(\frac{2b-1}{2}\right) = (-1)^b \frac{\binom{2b}{b}}{4^b} (1 - i)^b, \quad \text{hence} \quad I_b\left(\frac{2b-1}{2}\right) = (-1)^{b+1} \frac{\binom{2b}{b}}{4^b} \text{Im}(s^b),$$

of magnitude $2^{\lfloor b/2 \rfloor} \binom{2b}{b} / 4^b$, vanishing iff $4 \mid b$.

Proof. By Proposition ?? the lower parameter of g_b at $m = \frac{2b-1}{2}$ is $m - b + 1 = \frac{1}{2}$, and that of g_{b-1} is $\frac{3}{2}$. With $z = 1 - i$ (so $z^2 = -2i$, $1 + z = 2 - i$, $1 - z = i$) the quadratic transformations

$${}_2F_1\left(a, a + \frac{1}{2}; \frac{1}{2}; z^2\right) = \frac{1}{2}[(1 + z)^{-2a} + (1 - z)^{-2a}], \quad {}_2F_1\left(a, a + \frac{1}{2}; \frac{3}{2}; z^2\right) = \frac{(1 + z)^{1-2a} - (1 - z)^{1-2a}}{2z(1 - 2a)}$$

(at $a = -\frac{b}{2}$ and $a = -\frac{b-1}{2}$) give ${}_2F_1(\cdots; \frac{1}{2}; -2i) = \frac{1}{2}[(2-i)^b + i^b]$ and ${}_2F_1(\cdots; \frac{3}{2}; -2i) = \frac{(2-i)^b - i^b}{2(1-i)b}$. Using $\binom{(2b-1)/2}{b} = \binom{2b}{b}/4^b$ and $\binom{(2b-1)/2}{b-1} = 2b\binom{2b}{b}/4^b$, write $C = \binom{2b}{b}/4^b$; with $s/(1-i) = i$,

$$g_b = Cs^{b-1}(1+i)^{\frac{1}{2}}[(2-i)^b + i^b], \quad g_{b-1} = Cs^{b-1}(1+i)^{\frac{1}{2}}[(2-i)^b - i^b].$$

Subtracting, $(2-i)^b$ cancels: $G_b = Cs^{b-1}(1+i)i^b = C(si)^b = (-1)^b C(1-i)^b$, since $si = -(1-i)$. Finally $(1-i)^b = \overline{s^b}$, so $\text{Im } G_b = (-1)^{b+1} C \text{Im}(s^b)$ and $|\text{Im}(s^b)| = 2^{\lfloor b/2 \rfloor}$ for $b \not\equiv 0, 0$ for $4 \mid b$. $\square$

Remark 1. Theorem ?? unifies both regimes: for $4 \mid b$ it gives the rational root $\frac{2b-1}{2}$ of Q_b ; for $b \equiv 2, 3$ it shows the nearest half-integer is never a root, with an explicit central-binomial value. (Verified symbolically $b \leq 15$; magnitudes are the OEIS A001790 numerators, $b \leq 23$.)

3 Arithmetic, not analytic

Proposition 3. *For every $b \equiv 2, 3 \pmod{4}$, $b \geq 6$, the real polynomial I_b has a zero in (b, cb^2) , $c \approx 0.33$.*

Proof. $I_b(b)$ and $I_b(M)$ (for M past the largest root, where $\text{sgn } I_b = \text{sgn}(\text{Im } s^b/b!)$) differ in sign for these b ; a sign change at integer arguments $\geq b$ forces a real root. $\square$

Hence no uniform analytic lower bound $|I_b(m)| > 0$ can hold for real $m \geq b$: any log-concavity / leading-term dominance argument fails exactly where the in-range real roots live, so it can only reach $m \gtrsim cb^2$, no better than brute force. The genuine content of $(\star)$ — that the irrational in-range roots avoid the integers — is *arithmetic*. This retires the planned log-concavity route.

4 Frontier

Theorem 2. *For every $b \equiv 2, 3 \pmod{4}$, $3 \leq b \leq 70$, the only rational roots of I_b are integers $< b$; thus Q_b has no integer root $\geq b$ and the law holds. Each Q_b is irreducible over $\mathbb{Q}$. (Exact rational-root enumeration; supplementary $I_b(m) \neq 0$ for all integers $m \in [b, b^2]$, $b \leq 150$.)*

The gap

$(\star)$ for $b \equiv 2, 3$ is open: Q_b has no integer root $\geq b$ (empirically irreducible). Dead, structurally: single-prime Newton, Eisenstein, 2-adic truncation, covering sets (06-08 CODE), and now analytic dominance (Prop. ??). Live, global: uniform irreducibility of the Gegenbauer-parameter family $C_b^{(-m)}(-(1+i)/2)$; a Legendre $(A^{-1/2} = \sum P_n(-s/2)u^n)$ reflection identity; or a family-coefficient growth/Perron bound on rational roots.